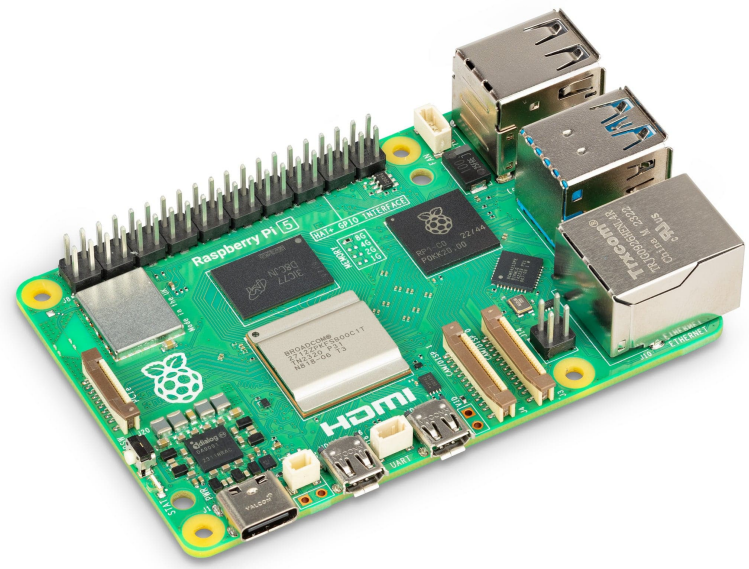
Lab 6: LLM with Ollama

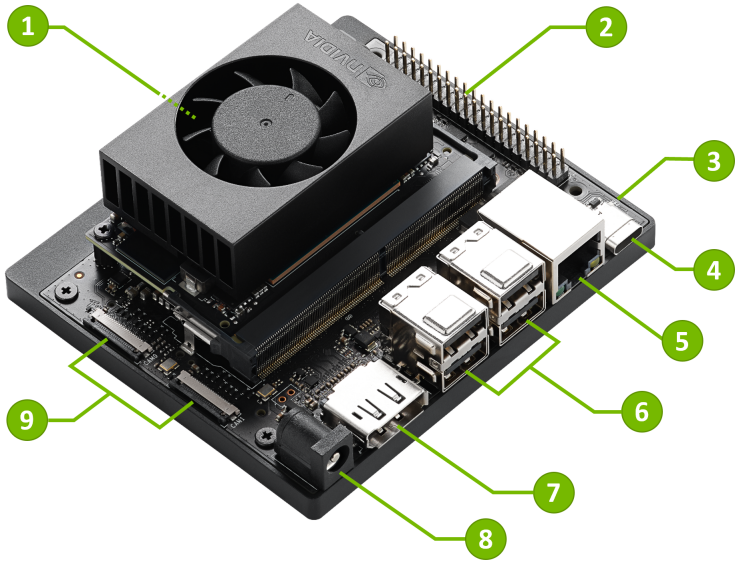
Raspberry Pi 5

The Raspberry Pi 5 is a powerful, affordable single-board computer featuring a quad-core Arm CPU.



Nvidia Jetson Orin Nano

The NVIDIA Jetson Orin Nano Super Developer Kit delivers 67 INT8 TOPS of AI performance, powered by an Ampere GPU with 1024 CUDA cores and a 6-core Arm Cortex-A78AE CPU. It features 8GB LPDDR5 memory and operates within a 7W–25W power envelope.



Nvidia Jetson AGX Orin

The NVIDIA Jetson AGX Orin 64GB Developer Kit delivers 275 INT8 TOPS of AI performance, powered by an Ampere GPU with 2048 CUDA cores, 64 tensor cores, and a 12-core Arm Cortex-A78AE CPU. It features 64GB LPDDR5 memory and operates within a 15W–60W power envelope.

SSH into device

Raspberry Pi 5:

```
$ ssh raspberrypi@192.168.0.XXX
password: raspberrypi
```

Jetson Orin Nano / AGX Orin:

```
$ ssh ollama@192.168.0.XXX
password: ollama
```

What is ollama and LLaMA

ollama

- Open-source LLM inference framework
- Run models locally on your own hardware
- Supports Llama 3.3, DeepSeek-R1, Phi-4, Mistral, Gemma 3, and more
- Available for macOS, Linux, and Windows
- Simple CLI: `ollama run <model> --verbose`

LLaMA

- Large Language Model Meta AI
- Open-source model family by Meta
- Available in various sizes (1B, 3B, 8B, 70B, 405B)
- Optimized for both research and commercial use

Meta

Run ollama llama3.2:1b on Raspberrypi

```
$ source ~/ollama/bin/activate
$ ollama run llama3.2:1b --verbose
```

```
>>> can you describe Diriyah in Riyadh briefly?
Diriyah is a historic city located in the Kingdom of
Saudi Arabia, which serves as the country's first
capital. It is situated about 30 kilometers south of...

total duration: 19.668060161s
load duration: 43.290122ms
prompt eval count: 34 token(s)
prompt eval duration: 284.937688ms
prompt eval rate: 119.32 tokens/s
eval count: 153 token(s)
eval duration: 19.339131658s
eval rate: 7.91 tokens/s
```

Metric

Metric	Meaning	Description
Total Duration (2.620s)	Full execution time	Total time from starting the command to completing the response, including model loading, input processing, and output generation.
Load Duration (39.95ms)	Model loading time	Time taken to load the model or necessary components into memory.
Prompt Eval Count (32 tokens)	Number of input tokens	Number of tokens the model evaluated from the input prompt.
Prompt Eval Duration (1.645s)	Input processing time	Time the model spent understanding and analyzing the input prompt.
Prompt Eval Rate (19.46 tokens/s)	Input processing speed	Speed at which the model processes tokens during prompt evaluation.

Metric

Metric	Meaning	Description
Eval Count (8 tokens)	Number of output tokens	Number of tokens generated in the response (e.g., “The capital of France is Paris.” yields 8 tokens).
Eval Duration (889.94ms)	Output generation time	Time required to generate the model’s response after evaluating the input.
Eval Rate (8.99 tokens/s)	Output generation speed	Rate at which the model generates tokens during response generation. Higher values indicate faster output production.

Performance Comparison: llama3.2:1b

Metric	RPi 5	Orin Nano	AGX Orin	A100
Total Duration	19.668s	5.166s	2.622s	0.789s
Load Duration	43.29ms	247.08ms	156.96ms	147.85ms
Prompt Eval Count	34 tokens	34 tokens	34 tokens	34 tokens
Prompt Eval Duration	284.94ms	35.30ms	18.63ms	4.70ms
Prompt Eval Rate	119.32 t/s	963.17 t/s	1825.23 t/s	7232.17 t/s
Eval Count	153 tokens	205 tokens	209 tokens	226 tokens
Eval Duration	19.339s	4.549s	2.229s	0.478s
Eval Rate	7.91 t/s	45.07 t/s	93.75 t/s	472.89 t/s

Prompt: *“Can you describe Diriyah in Riyadh briefly?”*

LLM can make errors

```
>>> what is the straight-line distance from Riyadh
    to Jeddah?
The straight-line distance from Riyadh to Jeddah is
approximately 80 kilometers (50 miles).
```

Actual answer (from Google):

The shortest distance (air line) between Riyadh and Jeddah is **526.51 mi (847.33 km)**.

The shortest route between Riyadh and Jeddah is 593.76 mi (955.57 km).

Always verify LLM outputs!

Models Available on Devices

Model	Parameters	Size
`llama3.2`	1B	1.3 GB
`deepseek-r1`	1.5B	1.1 GB
`deepseek-r1`	7B	4.7 GB

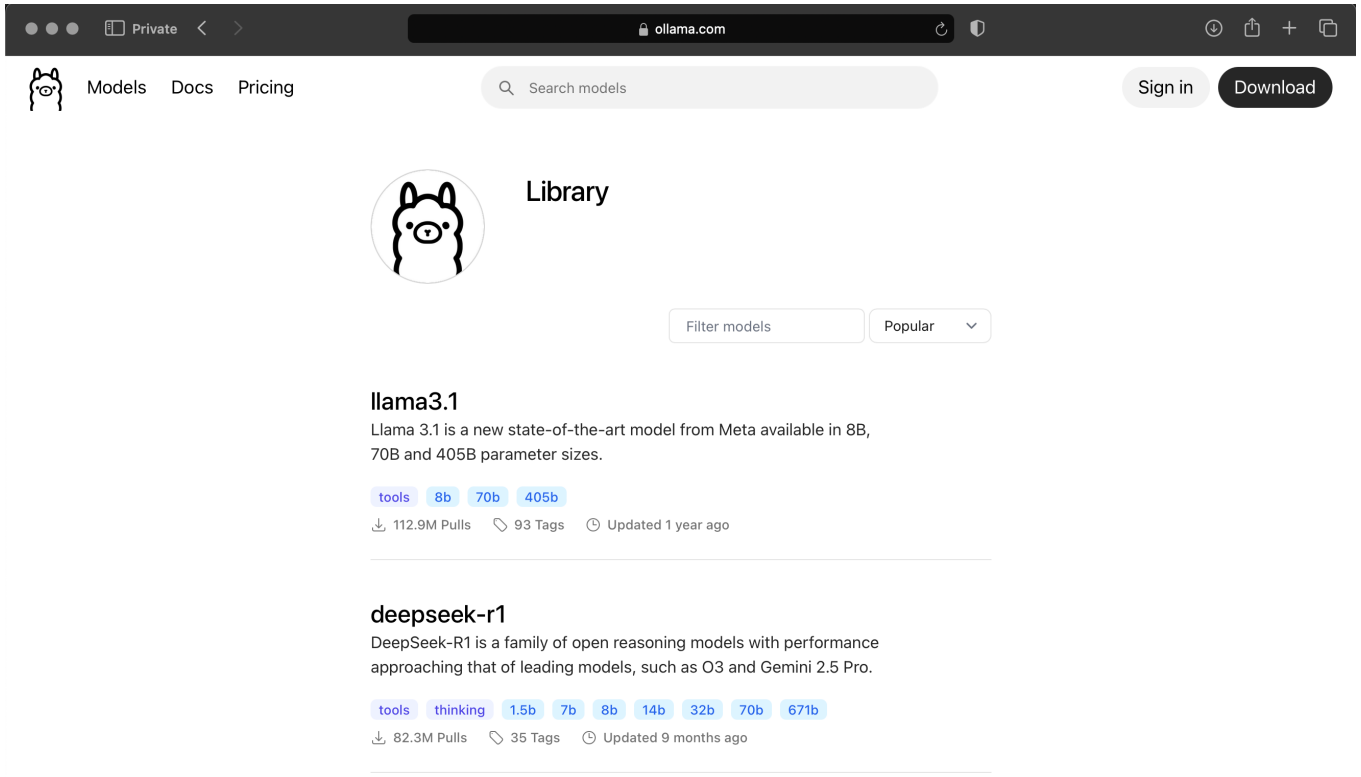
Commands:

```
$ ollama run llama3.2:1b --verbose
$ ollama run deepseek-r1:1.5b --verbose
$ ollama run deepseek-r1:7b --verbose
```

Explore More Models

Browse the full Ollama model library at:

`ollama.com/library`



Connect to WiFi

SSID	`TP-Link_BC8C`
Password	`default`

IP Pool – Raspberry Pi Devices

1. 192.168.0.165
2. 192.168.0.171
3. 192.168.0.181
4. 192.168.0.213
5. 192.168.0.147
6. 192.168.0.69
7. 192.168.0.103
8. 192.168.0.195
9. 192.168.0.53
10. 192.168.0.175
11. 192.168.0.196
12. 192.168.0.159
13. 192.168.0.178
14. 192.168.0.194
15. 192.168.0.36
16. 192.168.0.127
17. 192.168.0.158
18. 192.168.0.96
19. 192.168.0.200
20. 192.168.0.56

Username: `raspberrypi` Password: `raspberrypi`

IP Pool – Jetson & GPU Devices

Device	IP Address	Credentials
Orin Nano #1	192.168.0.201	ollama / ollama
Orin Nano #2	192.168.0.202	ollama / ollama
AGX Orin	192.168.0.205	ollama / ollama
A100 #1	192.168.0.203	ollama / ollama
A100 #2	192.168.0.204	ollama / ollama

SSH: `ssh ollama@<IP address>`